SWIPE-BASED RECRUITMENT SYSTEM WITH COMPATIBILITY ALGORITHM AND CAREER GUIDANCE

A DESIGN PROJECT-1 (5CS1990 / 5IT1990) REPORT

Submitted by

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BACHELOR OF TECHNOLOGY

In

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Under the Supervision of **Ms. Swetha C.B.**

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CERTIFICATE

This is to certify that the Design Project -1 work entitled "**SWIPE-BASED RECRUITMENT SYSTEM WITH COMPATIBILITY ALGORITHM AND CAREER GUIDANCE**" is the bonafide work done by , **RISHIT RAJAN SHUKLA (2023BCSE07AED244) , DEVANSH KHARE (2023BCSE07AED237)** , **ARYAVARDHAN SINGH SHEKHAWAT (2023BCSE07AED260)** submitted in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING** during the year **2025**.

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DECLARATION

This is to declare that the report titled "**SWIPE-BASED RECRUITMENT SYSTEM WITH COMPATIBILITY ALGORITHM AND CAREER GUIDANCE**" has been made for the partial fulfilment of the Programme - Bachelor of Technology in **Computer Science and Engineering**, under the guidance of **Ms. SWETHA C.B.** . I confirm that this report truly represents my work undertaken as a part of **my Design Project -1** work. This work is not a replication of work done previously by any other person. I also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the Mentor.

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- **Rishit Rajan Shukla**
- **Devansh Khare**
- **Aryavardhan Singh Shekhawat**

ABSTRACT

In these days's speedy-evolving virtual task marketplace, conventional recruitment portals like LinkedIn and Naukri depend upon key-word-based searches and static activity postings, which regularly fail to attach candidates with the maximum appropriate possibilities. the shortage of personalization, real-time interplay, and talent-based totally analysis effects in inefficiencies for each activity seekers and employers.

The proposed gadget, Swipe-based Recruitment machine with Compatibility set of rules and career guidance, pursuits to revolutionize recruitment using artificial Intelligence (AI), herbal Language Processing (NLP), and machine getting to know (ML). The cell-first platform introduces a swipe-based totally interface in which each employers and activity seekers can specific mutual interest, similar to the consumer experience of courting applications.

at the middle of the system is a compatibility set of rules that semantically fits resumes and task descriptions the usage of NLP and embedding-primarily based similarity scoring. on every occasion a candidate swipes proper on a employer, their resume is robotically shared with the company, streamlining the hiring workflow. For customers who are underqualified, the gadget presents profession steerage by using suggesting relevant gaining knowledge of resources and certifications that align with their favored job roles.

This AI-driven approach guarantees equity, performance, and engagement on each aspects of the hiring system. The project not simplest addresses obstacles in modern structures but additionally aligns with the modern staff's want for clever, customized, and talent-oriented recruitment.

LIST OF SYMBOLS & ABBREVIATIONS

Symbol / Term	Meaning
AI	Artificial Intelligence
NLP	Natural Language Processing
OCR	Optical Character Recognition
ML	Machine Learning
API	Application Programming Interface
SDK	Software Development Kit
UI / UX	User Interface / User Experience
DB	Database
GCP	Google Cloud Platform
ANN	Artificial Neural Network
LLM	Large Language Model

CONTENTS

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1. INTRODUCTION

In nowadays's digital generation, recruitment has transformed from paper-based totally hiring to online job portals. in spite of this evolution, inefficiencies persist — process seekers regularly observe to beside the point positions, and employers spend immoderate time filtering unsuitable candidates. The center problem lies within the lack of semantic information and mutual engagement among both aspects of the procedure.

The Swipe-based Recruitment device is developed to overcome those inefficiencies. This machine combines the electricity of AI-primarily based recommendation fashions with interactive design standards to provide a present day, clever, and attractive hiring enjoy. The app allows each employers and employees to swipe — "proper" for hobby and "left" for rejection — ensuring that fits occur handiest thru mutual consent.

The Compatibility algorithm acts as the selection-making middle. It uses NLP techniques which include tokenization, named entity recognition, and semantic embeddings (like BERT or Word2Vec) to evaluate resumes and activity descriptions. instead of key-word matching, it calculates a semantic similarity score, producing greater meaningful suits.

additionally, the profession steering characteristic guarantees inclusivity by suggesting upskilling sources, thus bridging the skill hole for aspiring applicants. thru this method, the venture creates a fair, shrewd, and adaptive hiring atmosphere that advantages each candidates and organizations.

2. LITERATURE REVIEW

PAPER 1: Embedding-Based Recommender System for Job to Candidate Matching on Scale

Author(s): X. Liu, Y. Wang, CareerBuilder Research Team

Date: 2021

Abstract: This paper introduces a large-scale candidate-job matching system using embeddings of resumes, job descriptions, and location data. A two-stage pipeline is employed: FAISS for fast retrieval and a reranking model to refine matches using contextual features such as skills, education, and work experience. The approach outperforms keyword-based systems by capturing deeper semantic relationships. The solution demonstrates scalability, operating efficiently on millions of resumes and job postings, while maintaining accuracy and speed.

Accomplishments:

1. Large-scale deployment with millions of records.
2. Semantic similarity improves over keyword matching.
3. Efficient real-time search via FAISS.

Limitations:

1. Requires advanced AI infrastructure.
2. Performance depends on resume data quality.
3. Limited transparency in embedding models.

Scope & Relevance: Directly relevant to our compatibility algorithm, which will also use embeddings for effective candidate-job matching.

PAPER 2: Fair Reciprocal Recommendation in Matching Markets.

Author(s): R. Burke, L. Xia, and J. Chen

Date: 2024

Abstract: This paper explores fairness in reciprocal recommendation systems such as dating and recruitment apps. It introduces envy-freeness and Nash Social Welfare (NSW) to ensure balanced opportunity distribution. The system accounts for the preferences of both parties in a two-sided market and reduces bias in the matching process. Experimental results demonstrate that fairness can be integrated without sacrificing overall match quality.

Accomplishments:

1. Introduces fairness metrics for reciprocal matching.
2. Maintains balance between performance and equity.
3. Supports mutual interest decision-making.

Limitations:

1. May lower efficiency under fairness constraints.
2. Computationally complex for large datasets.

3. Requires careful parameter tuning.

Scope & Relevance: Supports our swipe-based mutual interest model, ensuring fairness for both employers and candidates.

PAPER 3: AI-based Personalised Job Recommendation and Application Assistant Agent

Author(s): J. Sharma, F. Ali, and K. Wong

Date: 2023

Abstract: This work develops an AI assistant that helps candidates with job search, personalized recommendations, and application tracking. By analyzing resumes and job descriptions using NLP, it improves the relevance of recommendations and streamlines the application process. The assistant also enhances user engagement by reducing manual effort, making job searching more efficient and tailored.

Accomplishments:

1. End-to-end support for candidates.
2. Personalized recommendations with NLP.
3. Tracks applications and reduces effort.

Limitations:

1. Focused mainly on job seekers.
2. Limited features for employers.
3. Dependent on external job portals.

Scope & Relevance: Supports our career guidance module by highlighting personalized pathways for candidates.

PAPER 4: Resume Analyzer and Job Recommendation System

Author(s): P. Das, L. Green, and H. Zhao

Date: 2025

Abstract: This paper introduces an OCR and NLP-based system for parsing resumes and extracting details like skills, education, and work history. The extracted features are matched with job descriptions to provide recommendations. Additionally, it suggests certifications and training courses for underqualified candidates. The system demonstrates improved performance over keyword-based approaches.

Accomplishments:

1. Integrated resume parsing and job matching.
2. Provides upskilling suggestions.
3. Outperforms keyword-based systems.

Limitations:

1. OCR errors reduce accuracy.

2. Requires well-structured resumes.
3. Limited interactive features.

Scope & Relevance: Directly relevant to our app's career guidance feature for underqualified candidates.

PAPER 5: GIRL: Generative Job Recommendations with Large Language Models

Author(s): V. Singh, K. Tanaka, and L. Roberts

Date: 2023

Abstract: GIRL introduces a generative model for job recommendations using large language models (LLMs) and reinforcement learning. Instead of ranking jobs, the system generates tailored recommendations directly from candidate profiles. Experiments show superior personalization, although scalability remains a challenge due to resource demands.

Accomplishments:

1. Innovative use of LLMs for personalization.
2. Adapts to feedback using reinforcement learning.
3. Outperforms baseline models in personalization.

Limitations:

1. Very resource-intensive.
2. Early experimental stage.
3. Difficult to scale to mobile platforms.

Scope & Relevance: Highlights future potential of integrating LLM-driven personalization into our recruitment app.

3.IDENTIFICATION OF THE PROBLEM / SCOPE OF THE PROJECT

PROBLEM IDENTIFICATION:

1. According to current systems, searches are done based on keywords, which lacks the ability to provide contextual skill.
2. Existing job portals are unilateral as they do not provide a two-way interaction between the job seekers and the employers.
3. Absence of mobile first interactivity inhibits user interaction.
4. There was no career guidance or individualized learning advice integration.
5. Lack of AI-based flexibility, which is informed by human activity.

SCOPE:

The proposed next-generation recruitment platform in this project will be a combination of:

1. To match accurately based on AI-based compatibilities algorithms.
2. Swipe-based interaction model on mutual selection,
3. Career guidance module in professional development, and
4. Continuous improvement learning through feedback.

It strives to transform the hiring process into not only an automated one but also personalized, fair and interactive to the modern users.

4.PROPOSED IDEA / SOLUTION

The suggested system will provide a new way of recruiting with the application of AI, NLP, and interactions design to raise the attitudes of both candidates and employers.

- To create a mobile-first job search engine, which will enable candidates and organizations to interact with each other via a swipe-based interactive design, making the application of the employment process simple and fun.
- To apply an AI-based compatibility algorithm by analyzing of resumes and job descriptions in semantic mode with Natural Language Processing (NLP) and embedding models to guarantee meaningful and accurate matching.
- In order to allow mutual selection in a two-way swipe scheme, it is important to make sure that connections can be established only with the desire of the candidate and the employer.
- To incorporate automatic resume forwarding, where the candidate information is immediately sent to the employer when a successful match is achieved, less manual labour in recruitment.
- To develop a career guidance module that suggests personalized material on upskilling, courses, and certifications in order to adjust the underqualified candidates to the job requirements.
- To apply real-time feedback system of learning in which the user swipe information is used to constantly refine and better AI recommendation accuracy.
- To create a scalable backend architecture with Node.js and Firebase which will support many users in real time with low lattency and a high performance level.
- To have a fair, transparent, and flexible recruitment ecosystem, which will not only benefit the employer and job seeker by enhancing equality, intelligent and usability in hiring.

5.OBJECTIVES OF THE IDEA / SOLUTION AS DESIGN PROJECT

- To remove the distance between the candidates and the employers by creating a user-friendly recruitment application based on AI.
- To create a compatibility algorithm that will analyze both textual and contextual markers of resumes and job postings to provide more precise matches.
- To institute a system of matching between the two, where neither party is prejudiced against as both parties can make choices depending on preference and compatibility.
- To improve the user experience by providing a new mobile user-friendly interface that enables swiping to make hiring simpler.
- To incorporate career growth support in the app by recommending topical training programs and learning resources.
- To apply Machine Learning (ML) and NLP to predict, rank and match candidates and job roles in real time.
- To create a secure and efficient backend with node.js and Firebase to handle user authentication, data storage and real-time communication.
- To provide a contribution to the digital transformation of the recruitment with developing an intelligent, scalable, and adaptive solution to the contemporary job market.

6. PROCESS / METHODOLOGY

1. User & Data Layer (System Initialization)

- **Registration & Data Collection:** Candidates and Employers register, post jobs and upload resumes.
- **Resume Parsing:** NLP can be used to analyze the resume and retrieve the skills, education, and experience of the candidates.

2. AI Core Processing

- **Compatibility Algorithm:** Use AI (embeddings + contextual analysis) to match resumes with job roles.
- **Threshold Check: Inequality:** Customary applicants/jobs whose compatibility score is high with the swipe interface will be filtered out and then referred to career guidance direct underqualified applicants.

3. Interaction & Confirmation

- **Swipe Interaction:** Applicants and Employers swipe on suggested profiles/jobs.
- **Match and Exchange:** When a mutual right swipe is done, the match is automatically confirmed and a resume/exchange contact information is sent.

4. Value-Add & Optimization

- **Career Guidance:** Recommend courses and resources to underqualified candidates to be upskilled.
- **Feedback Loop:** Continuously optimise the AI recommendation model using all history swipe and successful match data.

7. REQUIREMENTS

Hardware Requirements

- CPU: Intel i5 or Ryzen 5 and above
- RAM: 16 GB minimum
- Storage: 512 GB SSD
- Mobile Devices: Android 10+ / iOS 15+

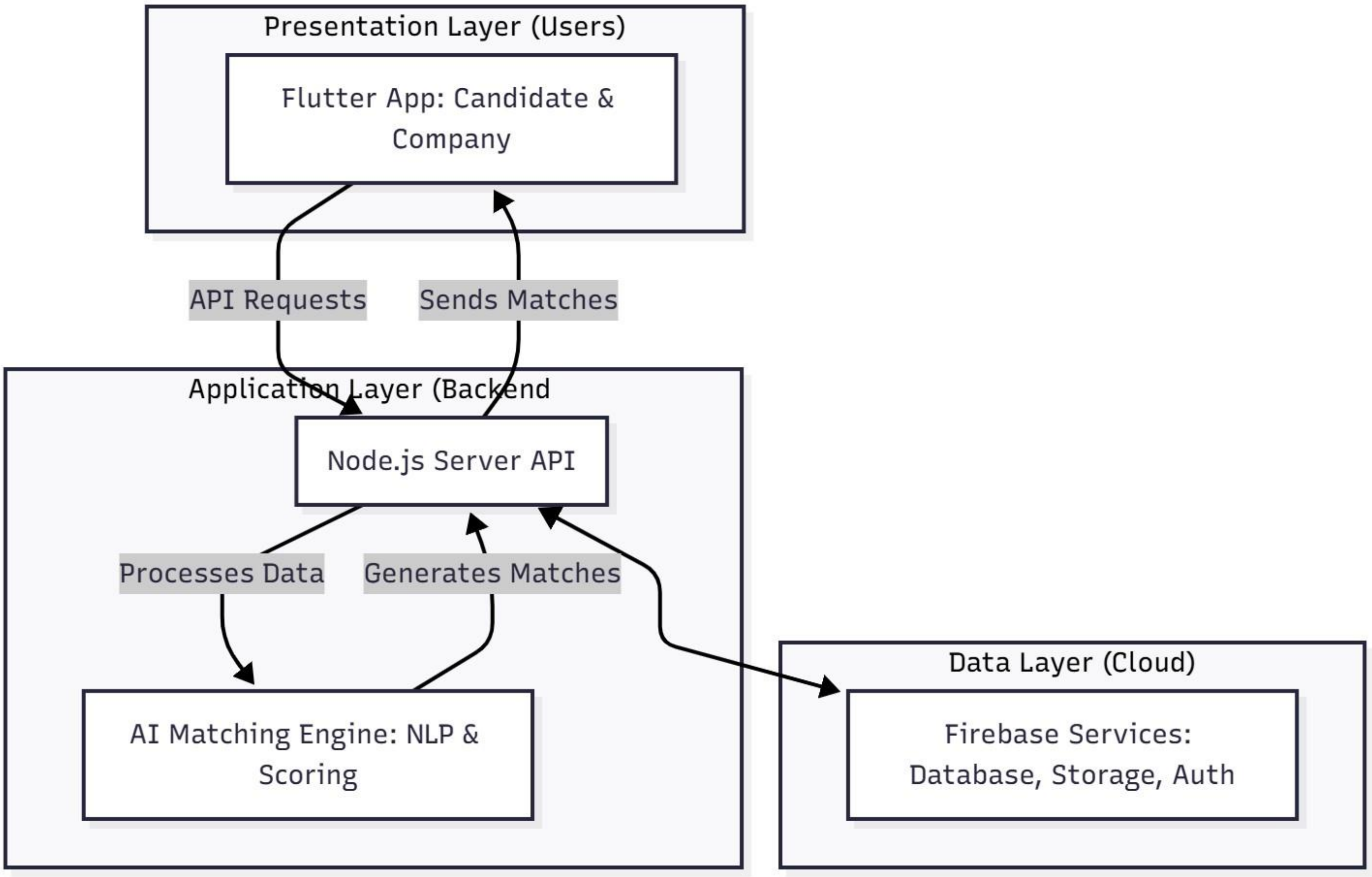
Software Requirements

- **Frontend:** Flutter SDK
- **Backend:** Node.js with Express.js
- **Database:** Firebase / MongoDB
- **AI Frameworks:** TensorFlow, NLTK, SpaCy
- **Development Tools:** Android Studio, VS Code, Git, Firebase Console

Dataset

- Resume Dataset (user-uploaded resumes)
- Job Dataset (company-provided postings)
- Skill Recommendation Dataset (for learning suggestions)

8. PROCESS / FLOW / DESIGN DIAGRAM



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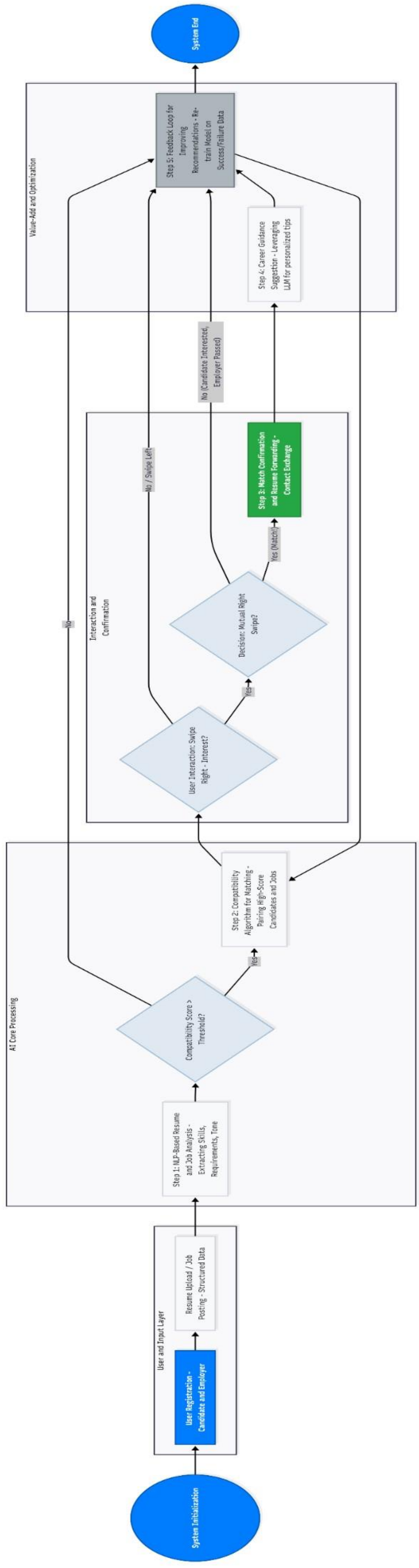

(METHADOLOGY FLOWCHART)

9. PROPOSED MODULES

Module Name	Description
User Module	Handles registration, login, and resume upload.
Employer Module	Allows posting jobs and viewing potential matches.
Compatibility Algorithm Module	Computes match scores using NLP and embeddings.
Swipe Module	Enables user interaction via swipe gestures.
Career Guidance Module	Suggests learning content and certifications.
Database Module	Stores user data, job listings, and match results.

10. NOVELTY OF THE IDEA / PROJECT

1. Swipe-based mutual matching: This has a mutual matching using a swipe mechanism, which allows both the employers and the candidates to show mutual interest, so that matches are done only where both parties concur.

2. AI compatibility algorithm: AI-powered and natural language processing models are used to match resumes and job descriptions to provide a higher quality and purposeful match to a candidate and a job than just a keyword filter.

3. Integrated skill instruction system: Recommends customized learning materials and credentials to enable underqualified applicants fulfill the demands of their preferred job positions.

4. Real-time feedback loop: This continuously using the analysis of swipes, choices, and patterns of engagement by users, the matching algorithm is improved to influence subsequent recommendations.

5. Mobile-first execution: It centers on creating a smoother, quicker and easier mobile experience among both Android and iOS users which will allow them to use the mobile platform at any time and location

11. CONCLUSION

The suggested Swipe-Based Recruitment System brings a new twist to the traditional recruitment system by creating a smart, AI-driven, and interactive recruitment platform that closes the divide between the job applicants and the employers. This system is a dynamic, mutual-matching search offering unlike traditional portals that match jobs based on stagnant job listing and manual search algorithms. It uses AI-enhanced compatibility scoring and NLP-enhanced resume parsing to make sure that the matches are relevant, as well as meaningful that is propelled by actual expertise and contextual knowledge instead of keywords.

The career guidance and skill development module is also an added value to the platform as it allows the underqualified candidates to upskill themselves to the positions that they want. This is because this system does not only help in making employment matches, but also makes personal and professional development.

Also, the real-time feedback learning will provide a self-enhancing ecosystem where each interaction refines future recommendations. The mobile-first design ensures accessibility and usability on various devices, which makes the system useful to users who are mobile at any time.

In general, this project can be seen as a stepping stone to next-generation recruitment ecosystems, that is, combining artificial intelligence, personalization based on the data, and the interactive experience within a unified system. This system can achieve a brilliant future in changing digital recruitment and empowering candidates by combining efficiency with reasonableness and individuality.

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